

Foundation Models in Single-Cell Biology

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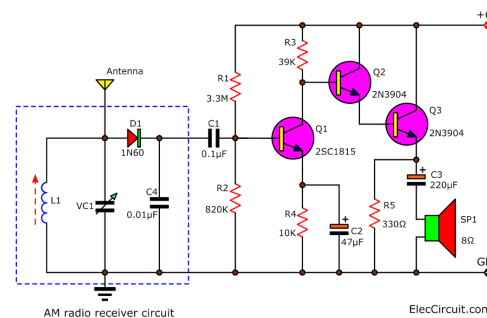
Foundation model era

➤ Machine learning paradigms

- **Feature engineering**: manual feature extraction (SVM, LightGBM, XGBoost)
- **Architecture engineering**: raw features, design deep network (CNN, LSTM)
- **Objective engineering**: pre-train large model and fine-tune it (ResNet50, Bert)
- **Prompt engineering**: prompt **foundation model** in zero/few shots

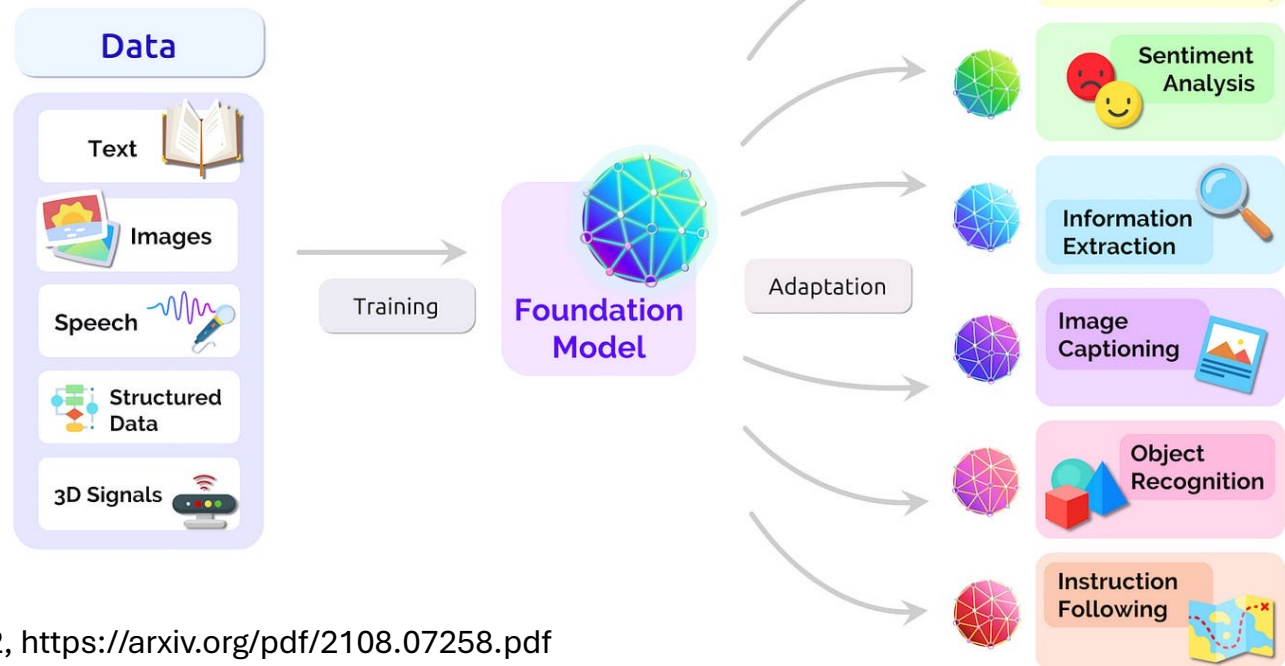
➤ Industrial era of artificial intelligence

- Text to text (GPT)
- Text to image (DALL-E)
- Connecting text and images (CLIP)
- Image segmentation (SAM)
- Protein sequence (ESM)



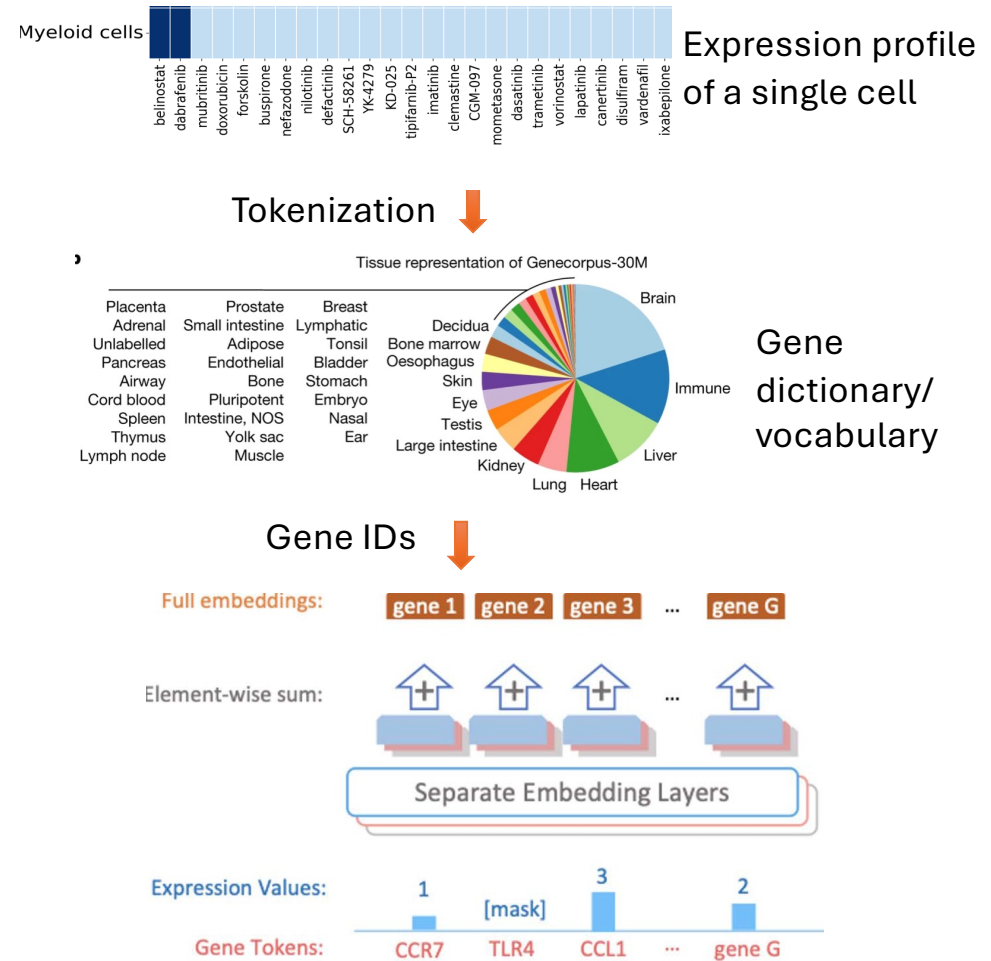
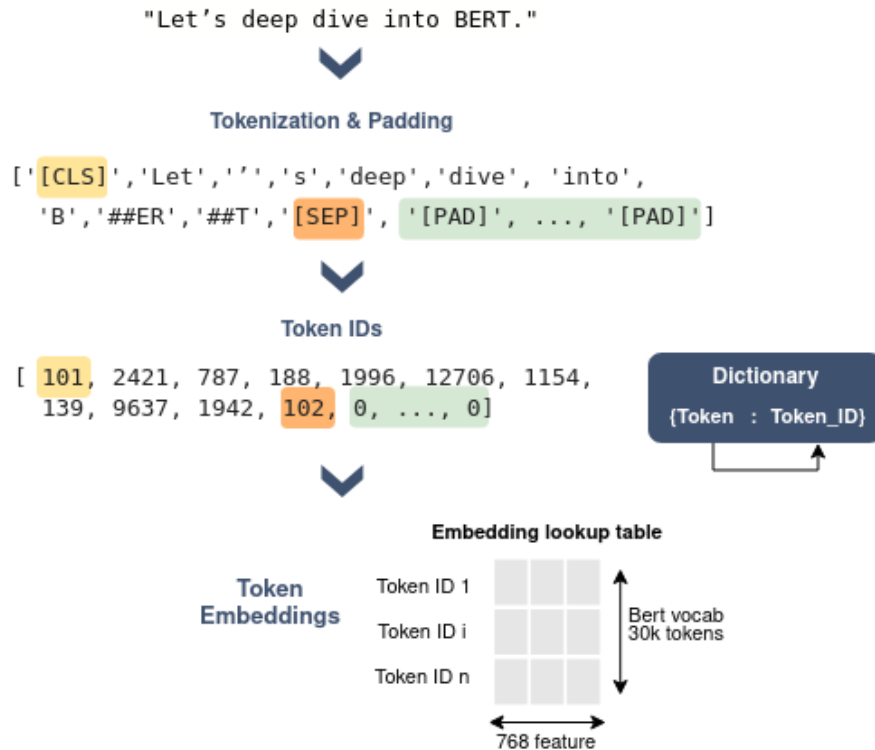
Foundation models

- Promptable
- Adaptable to many downstream tasks
- Emerging intelligence (trained on huge data)



Bommasani et al, 2022, <https://arxiv.org/pdf/2108.07258.pdf>

From LLM to scLLM



Representative Single-cell LLMs

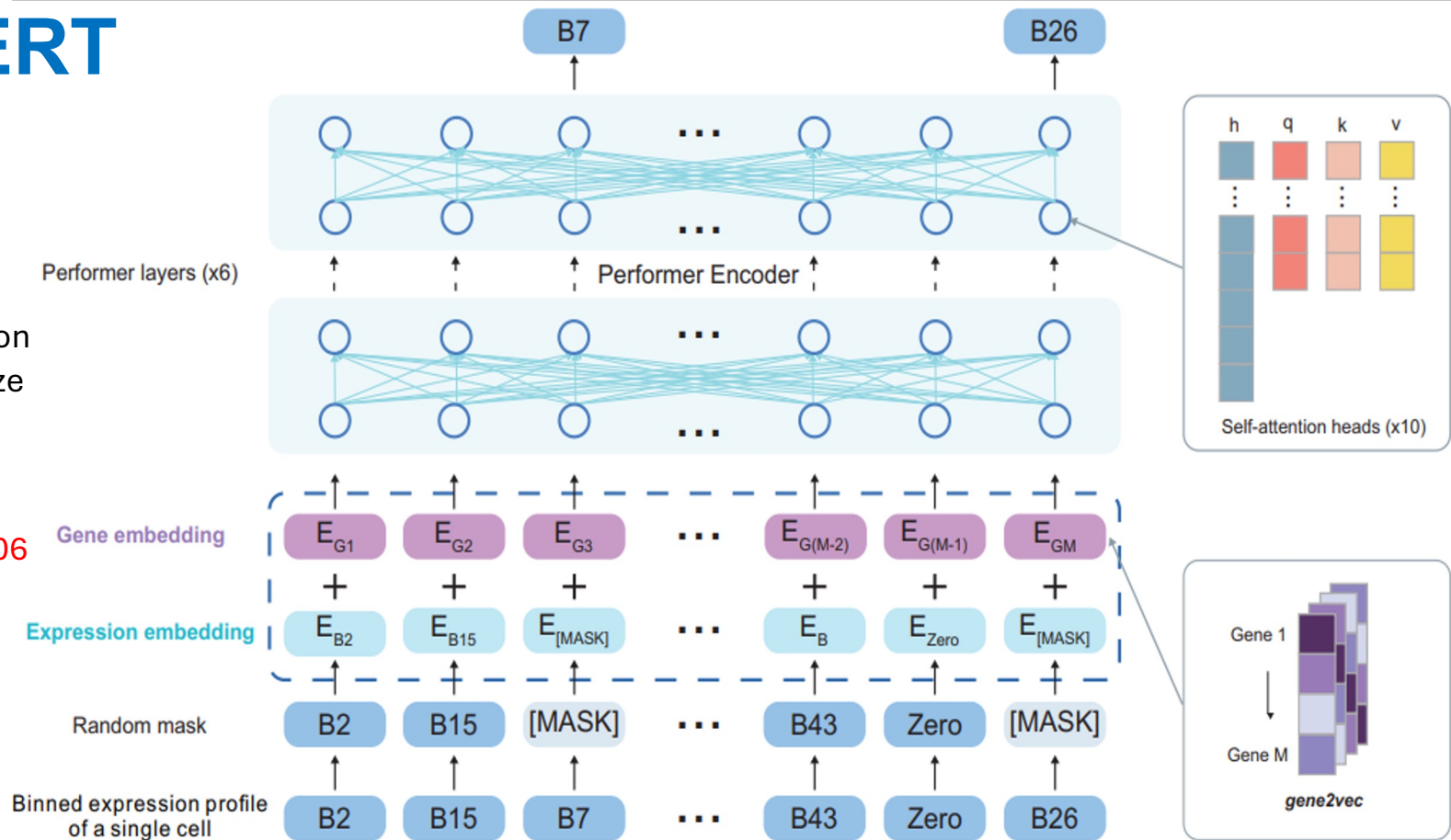
<i>Model</i>	<i>Paper</i>	<i># Parameters</i>	<i>Architecture</i>	<i>Training Data</i>	<i>Downstream Tasks</i>
scGPT	Cui et al. bioRxiv 2023	51 million	autoregressive transformer	33 million normal human cells (51 tissues, 441 studies)	cell-type annotation, multi- batch integration, multi- omic integration, perturbation prediction, and GRN inference
scBERT	Yang et al. Nature MI 2022	5 million	Performer (allowing for longer inputs)	209 human single-cell datasets comprising 74 tissues with 1M+ cells	cell type annotation
Geneformer	Theodoris et al. Nature 2023	40 million	Transformer	Genecorpus-30M- 29.9 million human single- cell transcriptomes	gene dosage sensitivity, chromatin, network dynamics,
scFoundation	Hao et al. bioRxiv 2023	100 million	Transformer (w/ trick to reduce # words)	50 million human cells (100+ tissue types, normal and disease)	clustering, perturbation prediction, drug response

Slide credit: Jian Ma (www.cs.cmu.edu/~jianma/talks/LLM-CGSI-slides-public.pdf)

scBERT

Binning expression profile to vectorize initial expression embedding

Vocabulary: 16906

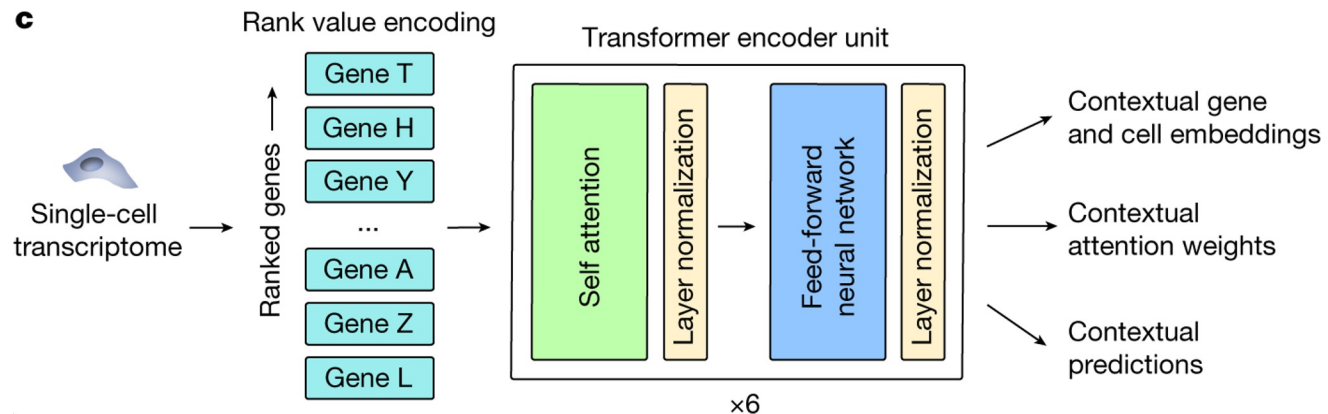
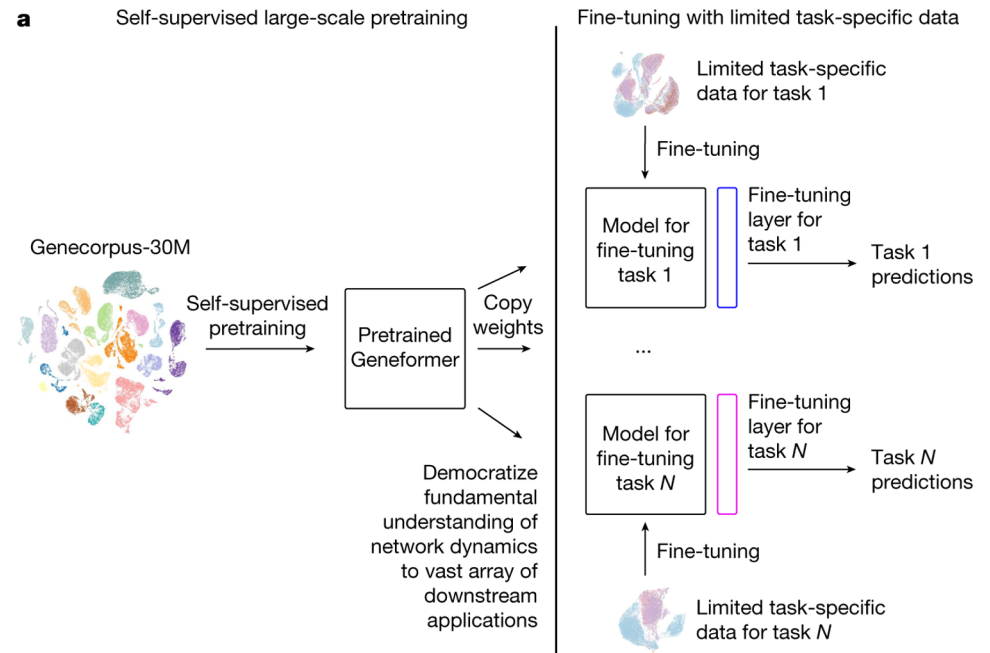


Train by PanglaoDB (209 human single-cell datasets consisting of **74 tissues**, and **1,126,580 cells**)

Yang, F., Wang, W., Wang, F. et al. "scBERT as a large-scale pretrained deep language model for cell type annotation of single-cell RNA-seq data". *Nat Mach Intell* (2022).

geneFormer

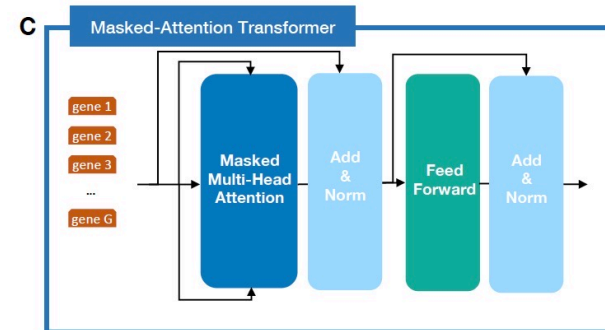
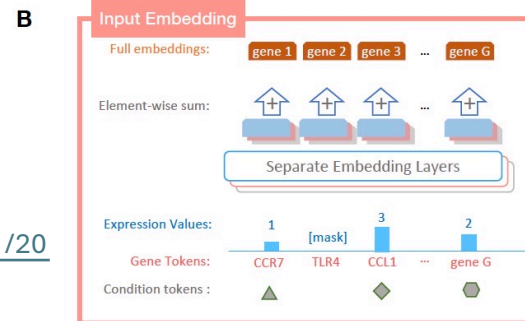
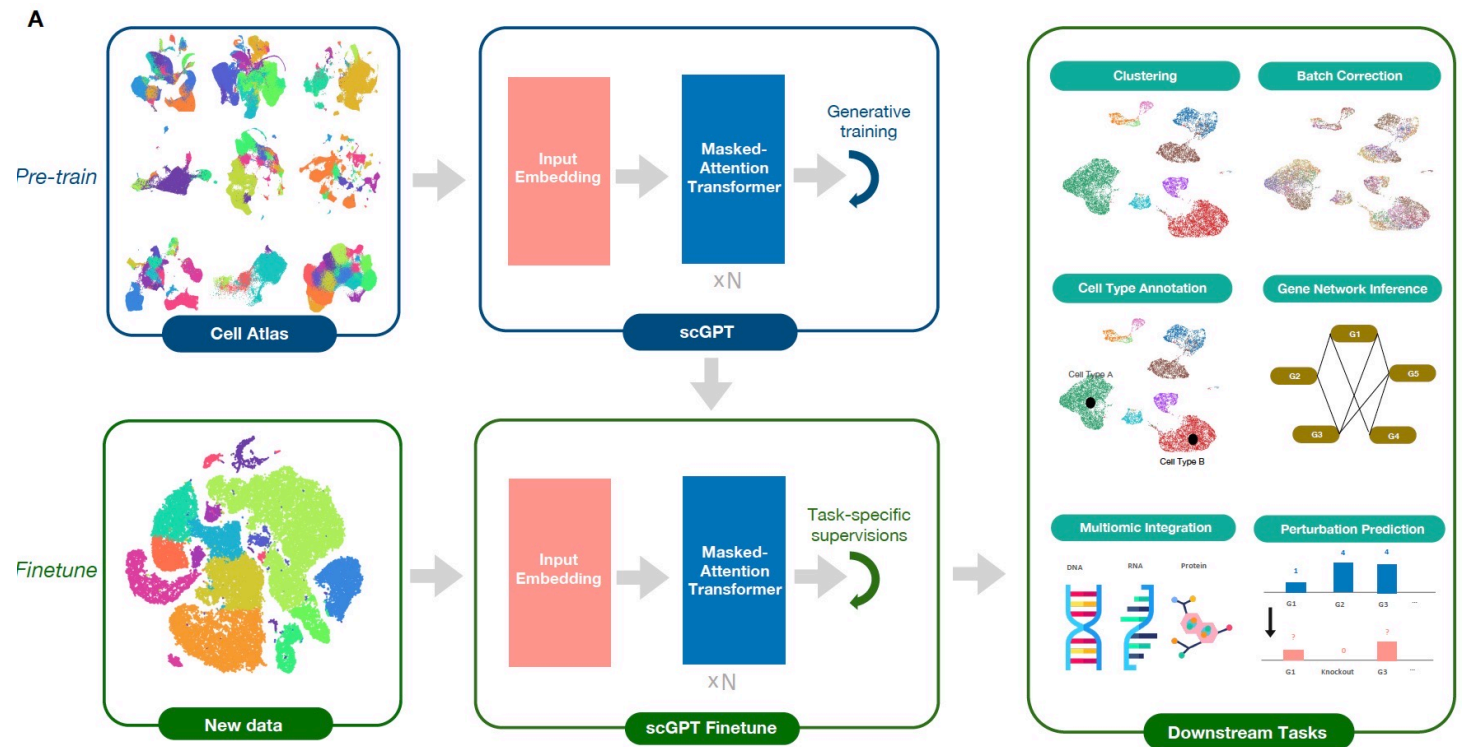
- ? Take raw count as input
- ? Embed expression on rank value only
- ? Average pool gene embedding to cell embedding
- ? Only published code at HuggingFace
- ? No documentation
- ? Vocabulary: 25424



Theodoris CV, Xiao L, Chopra A, Chaffin MD, Al Sayed ZR, Hill MC, Mantineo H, Brydon EM, Zeng Z, Liu XS, Ellinor PT. Transfer learning enables predictions in network biology. *Nature*. 2023 May 31:1-9.

scGPT

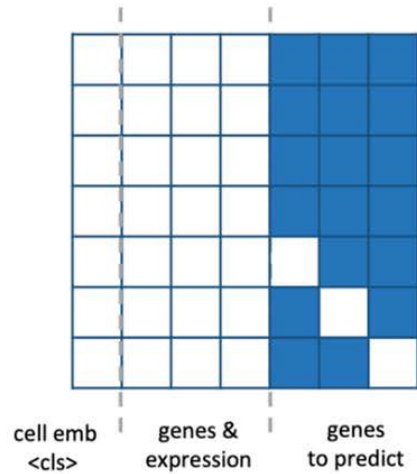
- Binning expression profile as well
- Set special token <CLS> token as cell embedding
- Fully open sourced and documented
- Vocabulary: 60697



Cui H, Wang C, Maan H, Pang K, Luo F, Wang B. scgpt: Towards building a foundation model for single-cell multi-omics using generative ai. bioRxiv. 2023 May
<https://www.biorxiv.org/content/10.1101/2023.04.30.538439v2.full.pdf>

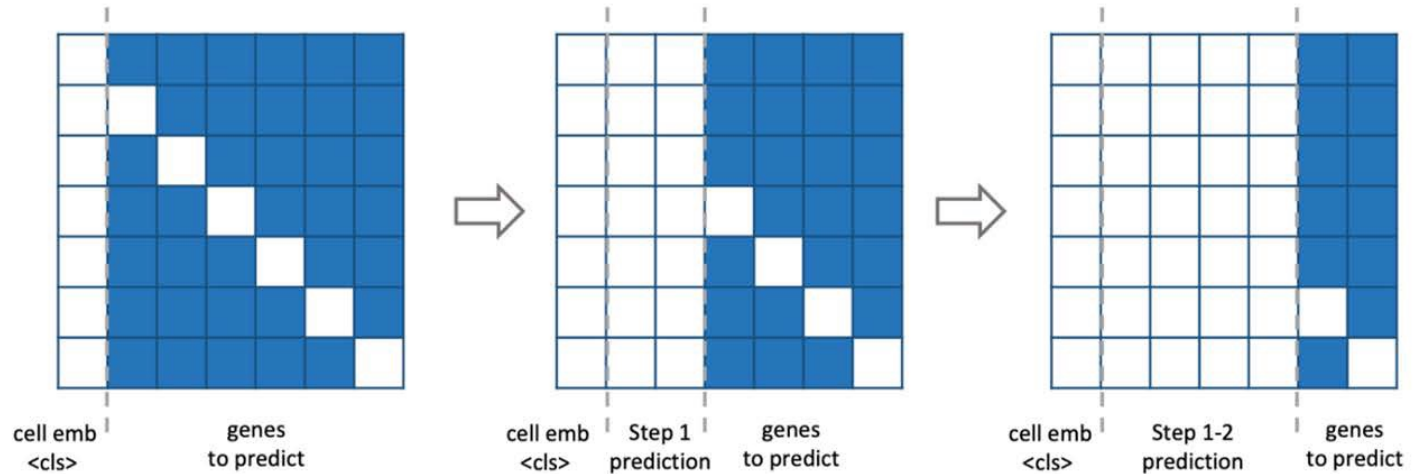
Generative pretraining and inference for scGPT

A Generative training



Teacher forcing training

B Generation steps during inference

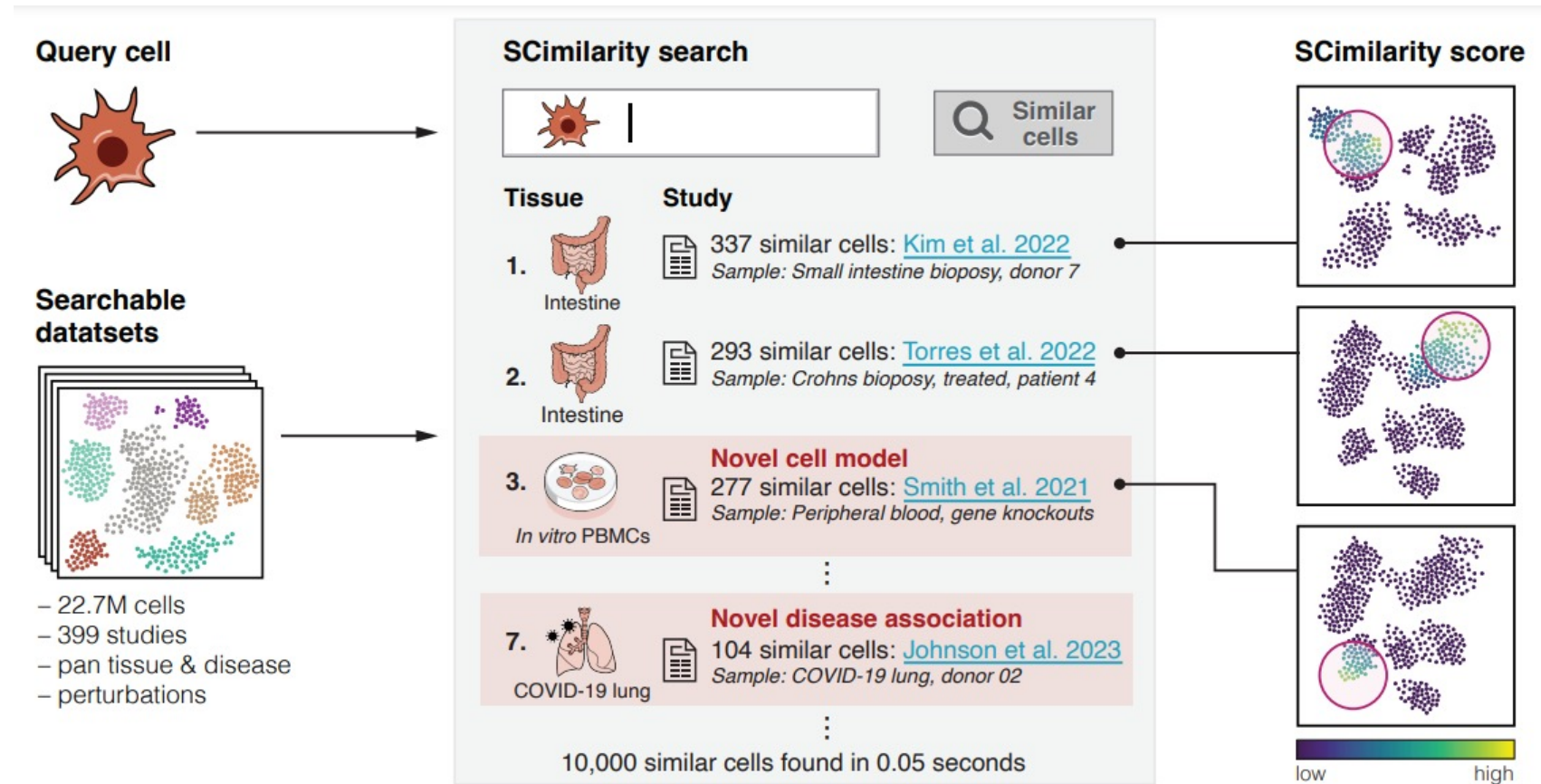


Step 1

Step 2

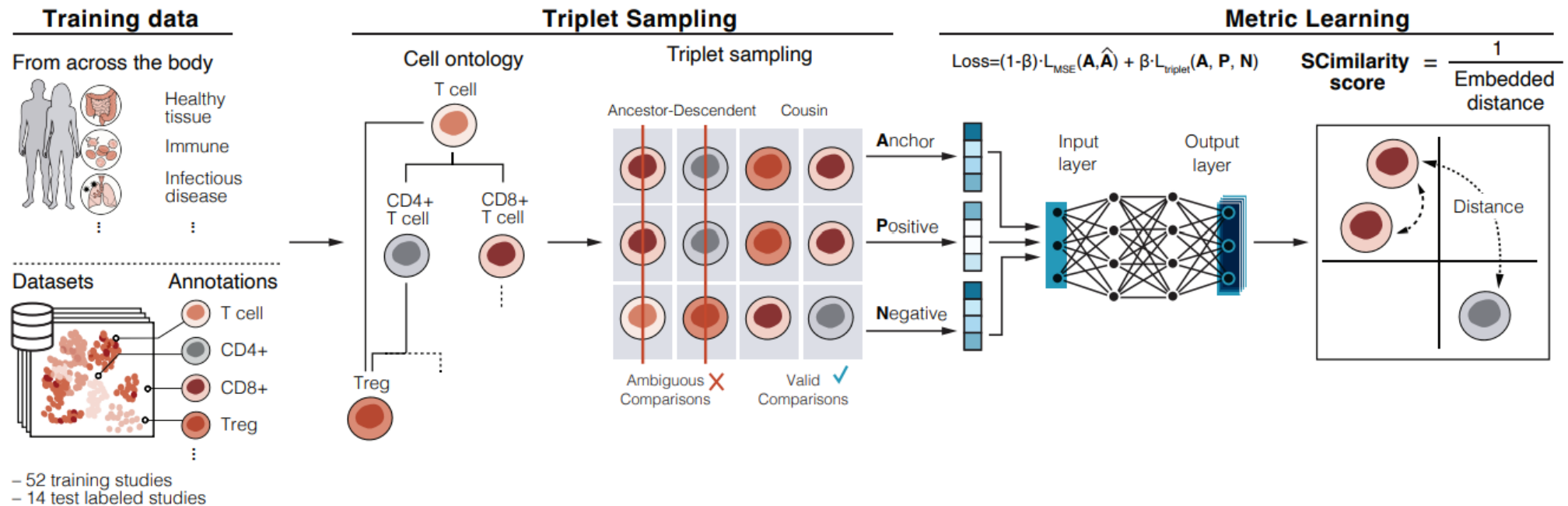
Step 3

SCimilarity



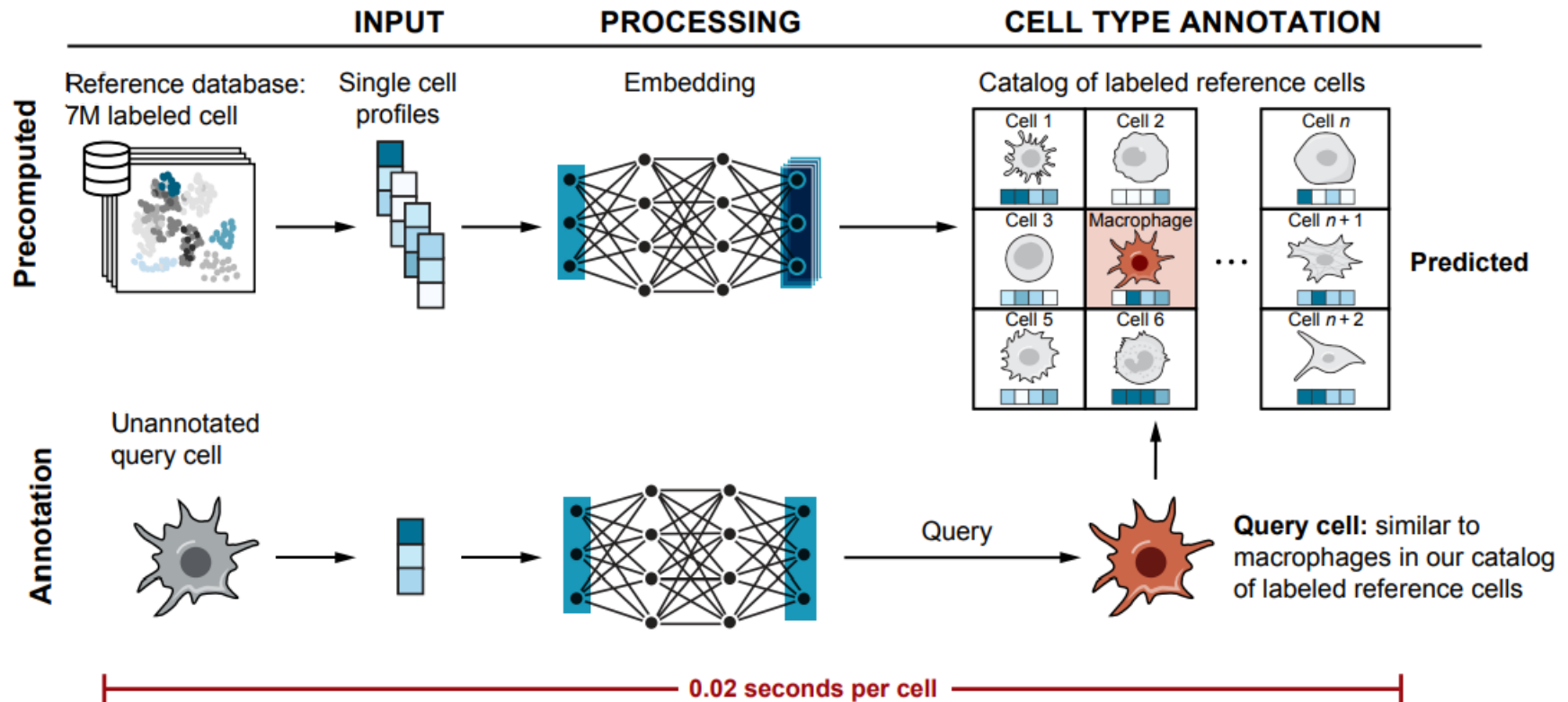
Heimberg G, Kuo T, DePianto D, ..., Aviv Regev. Scalable querying of human cell atlases via a foundational model reveals commonalities across fibrosis-associated macrophages[J]. bioRxiv, 2023: 2023.07. 18.549537.

Training for SCimilarity



Training SCimilarity models with a training set of **7,913,892** single-cell profiles from 52 studies

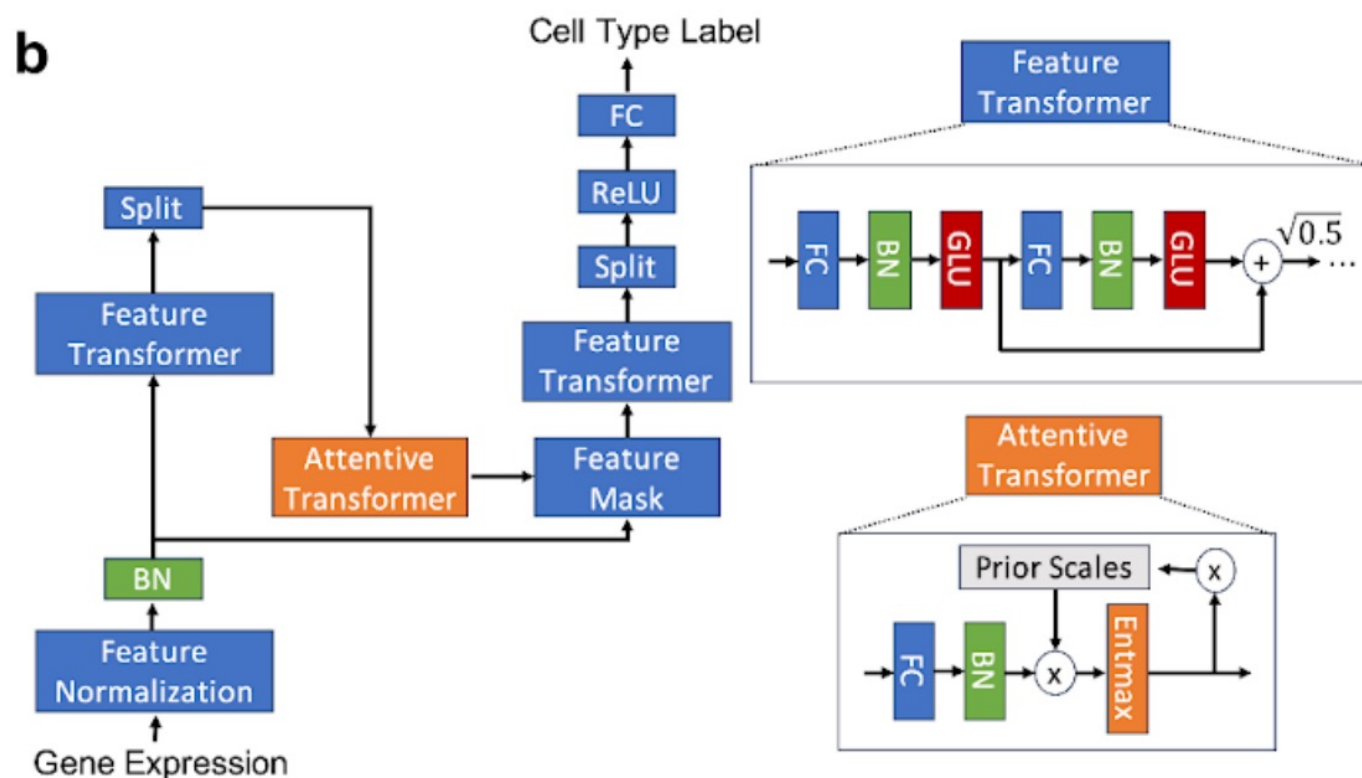
Cell type annotation for SCimilarity



- SCimilarity assigns an unannotated query cell to a cell type by finding similar cells in a labeled reference

scTab

- Based on the TabNet architecture
- The input to the model is all 19,331 protein-coding genes

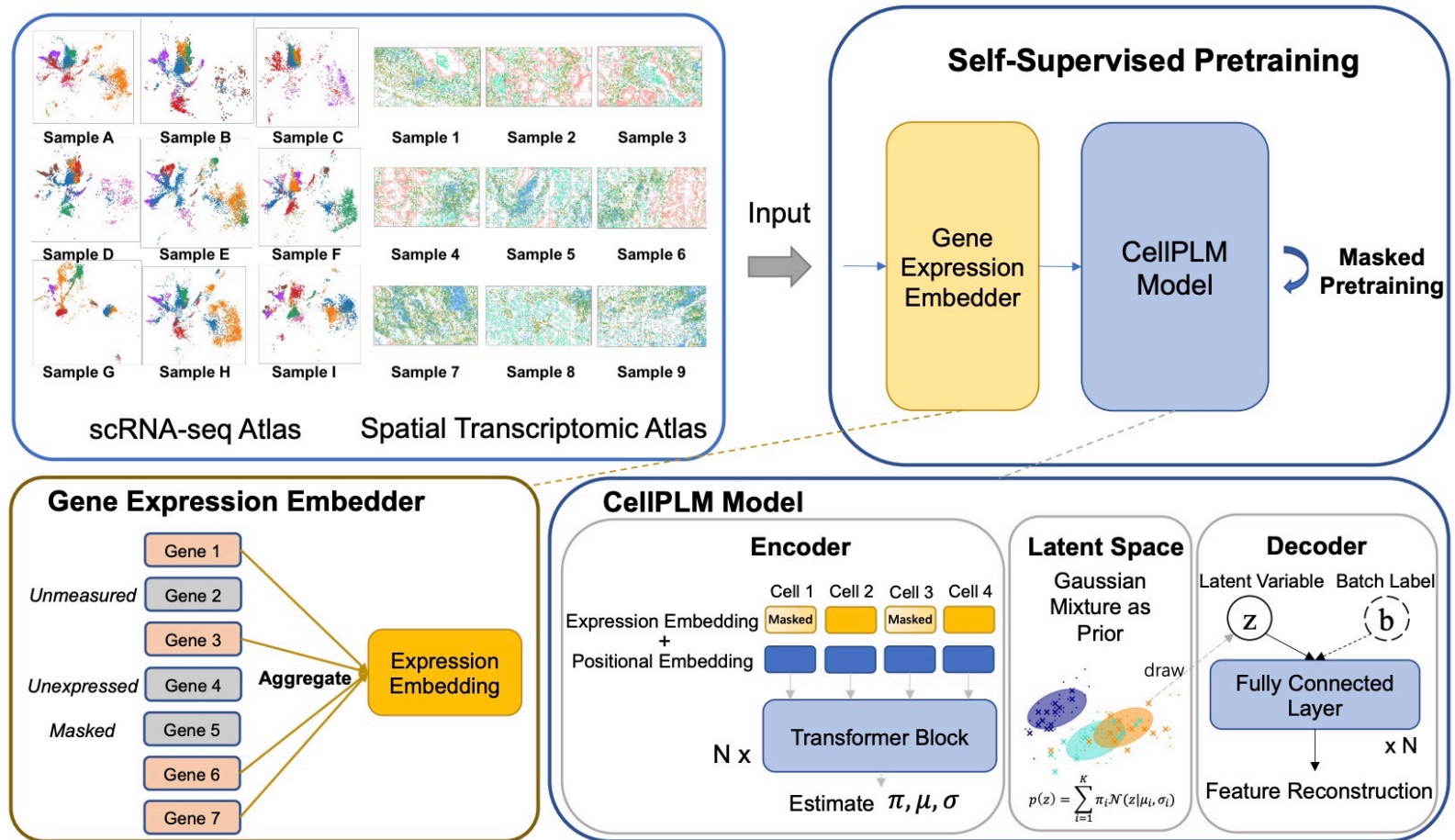


Train by 22.2 million cells from the assembled dataset span 5,052 unique donors, 249 datasets, **52 disease states**, **164 unique cell types**, and **56 different tissues**.

Fischer F, Fischer D S, Biederstedt E, Fabian J. Theis. Scaling cross-tissue single-cell annotation models[J]. bioRxiv, 2023.

cellPLM

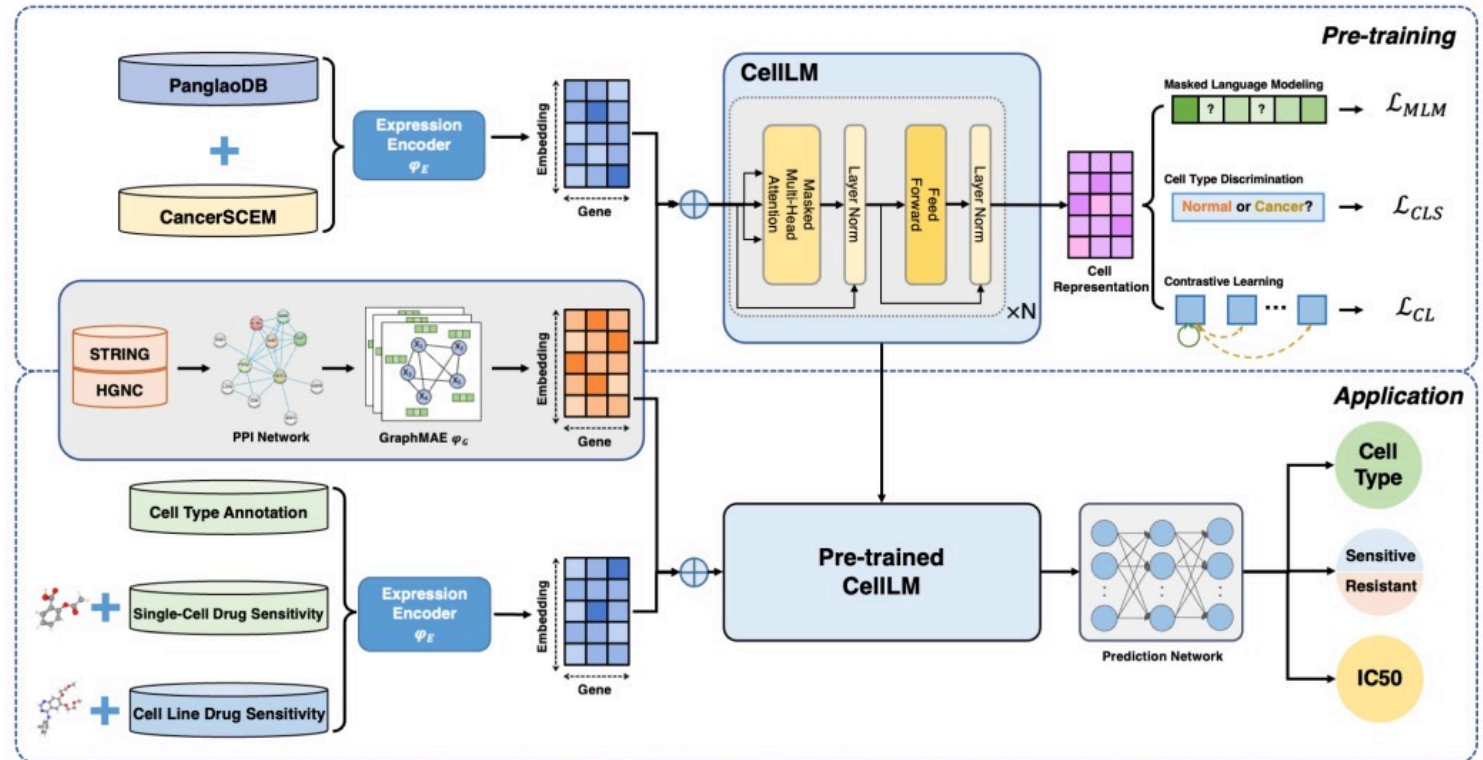
- Cell as token,
tissue as sentence
- Unsupervised MLM
(Masked Language
Model) pre-training



Wen, Tang, Dai, Ding, Jin, Xie, Jiliang Tang. CellPLM: Pre-training of Cell Language Model Beyond Single Cells. bioRxiv 2023.

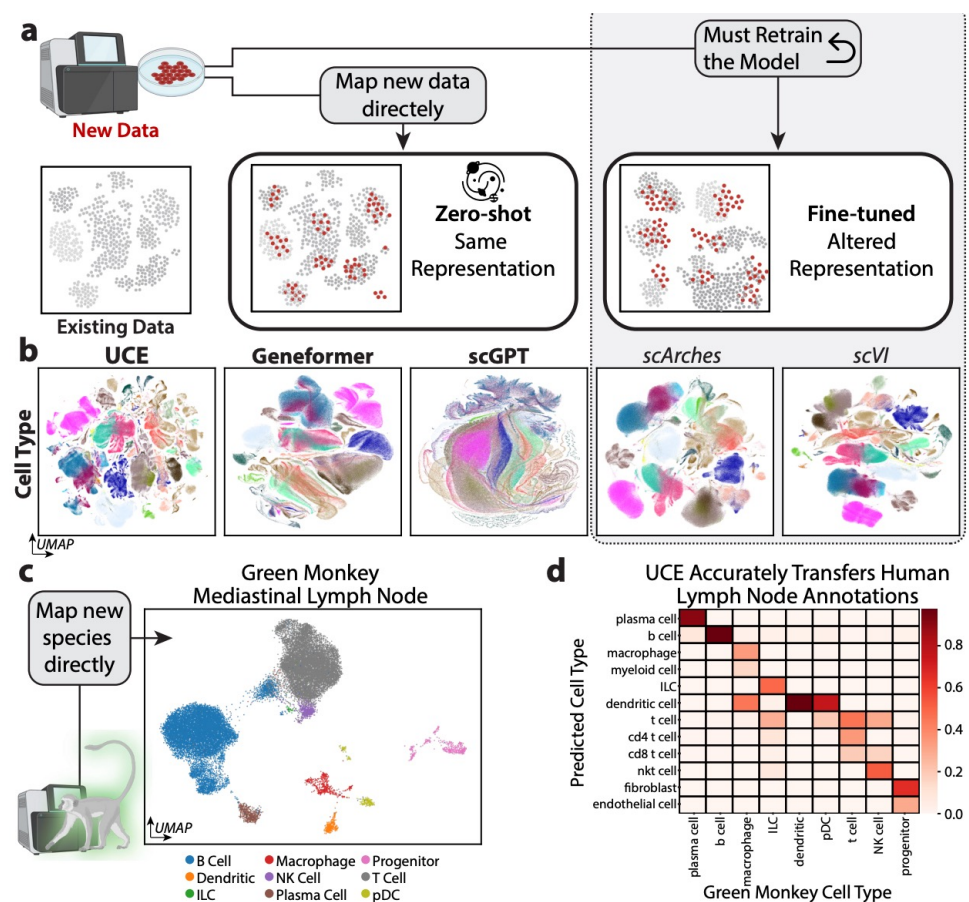
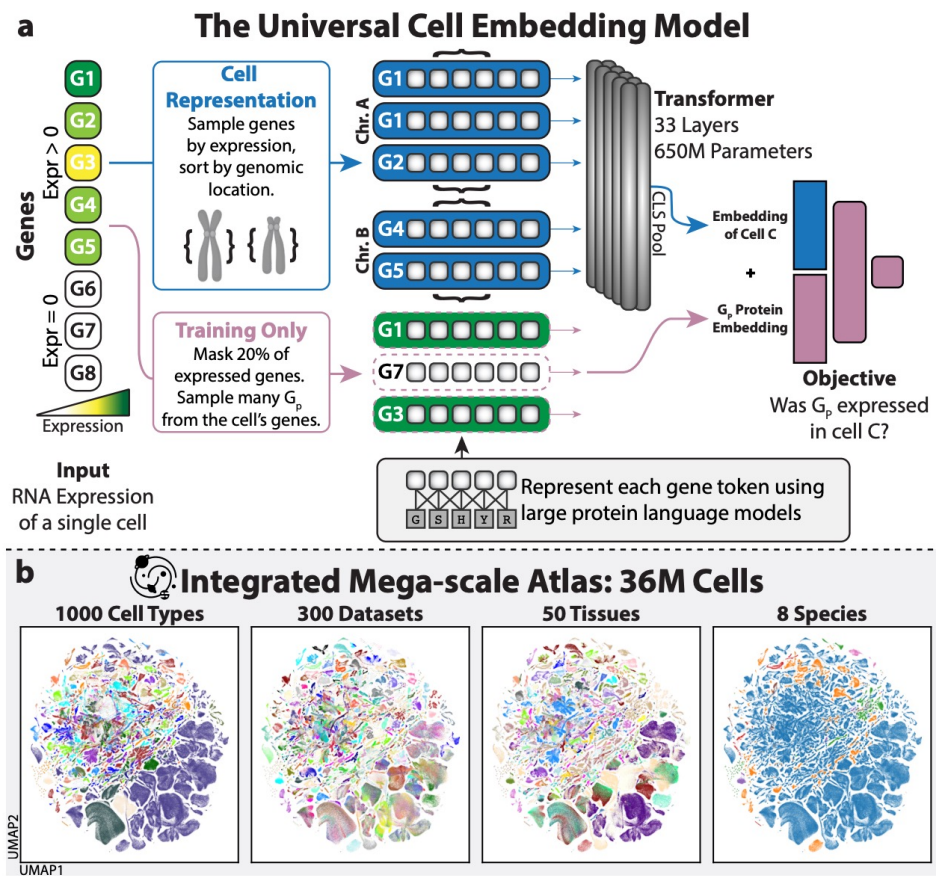
cellLM

- The gene expression data and gene interaction data are integrated
- Three self-supervised tasks : MLM, cell type discrimination, and contrastive learning
- Application: cell type annotation and drug sensitivity tasks at single-cell and cell line data levels.



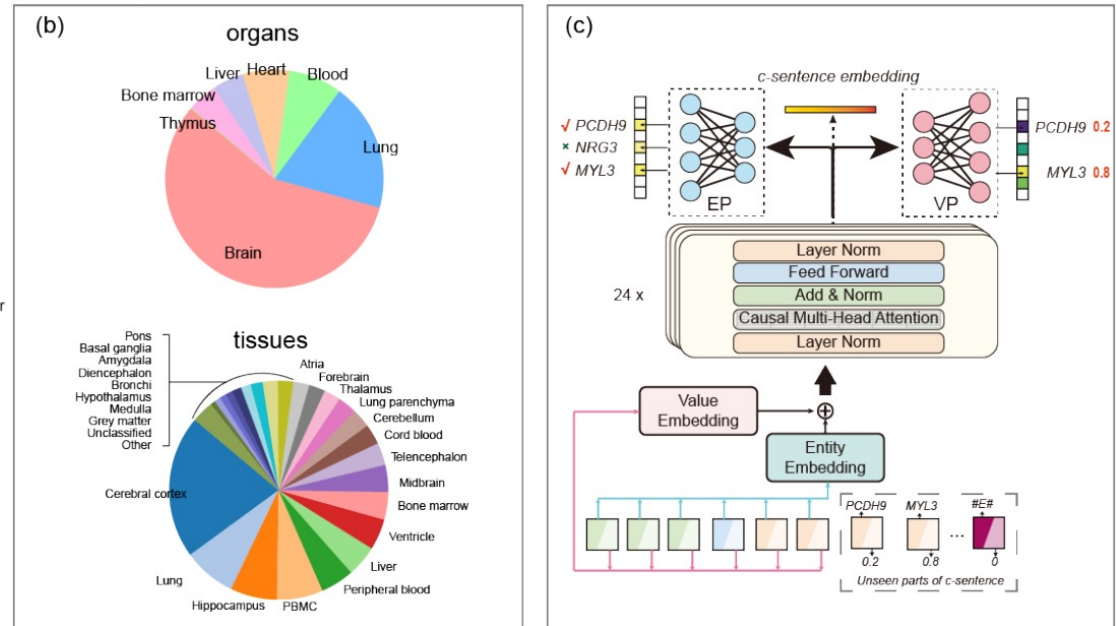
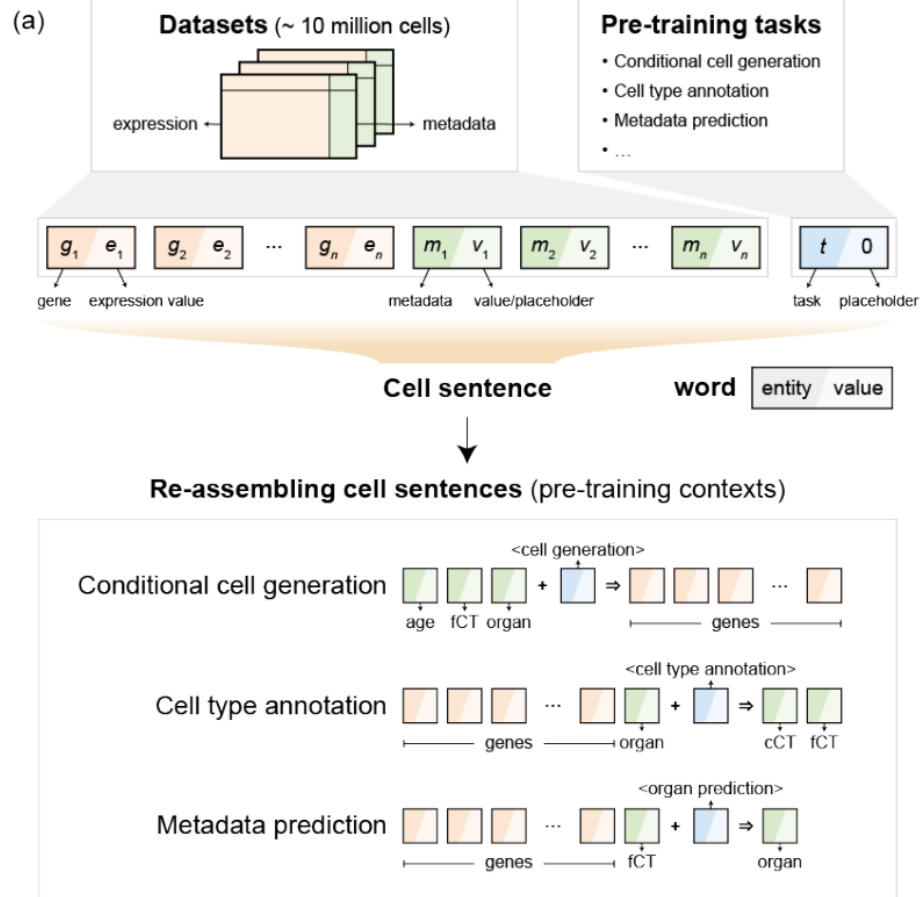
Zhao, Suyuan, Jiahuan Zhang, and Zaiqing Nie. "Large-scale cell representation learning via divide-and-conquer contrastive learning." arXiv preprint arXiv:2306.04371 (2023).

Universal Cell Embedding



Yanay , Yusuf , Ayush , Leon , Tabula, Stephen, Jure Leskovec. Universal Cell Embeddings: A Foundation Model for Cell Biology .
bioRxiv 2023.11.28.568918

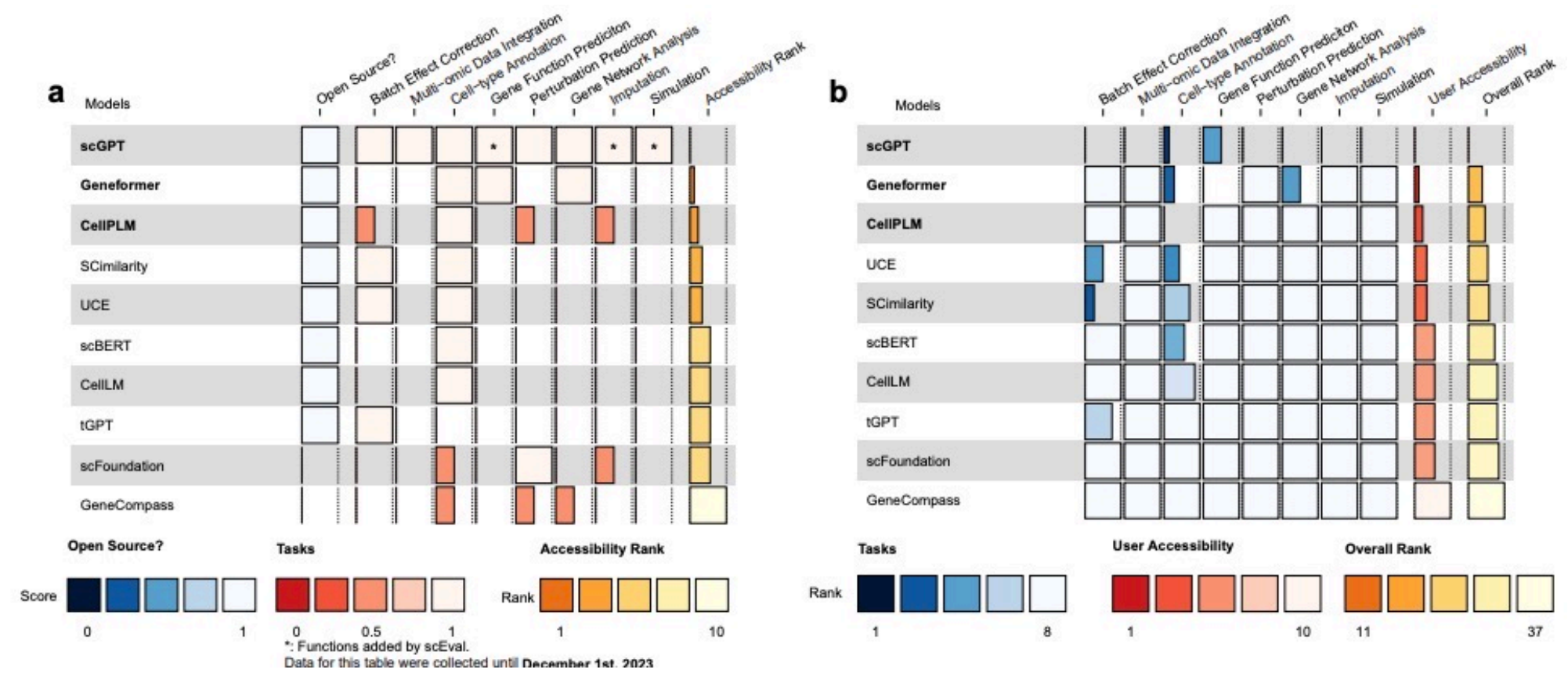
scMulan



- Binning expression profile into 10 equidistant bins
- Pretrained on 10M human single-cell transcriptomes
- scMulan is built upon a Transformer decoder-only architecture
- Gene token vocab size is 2000, cell type vocab size is 114

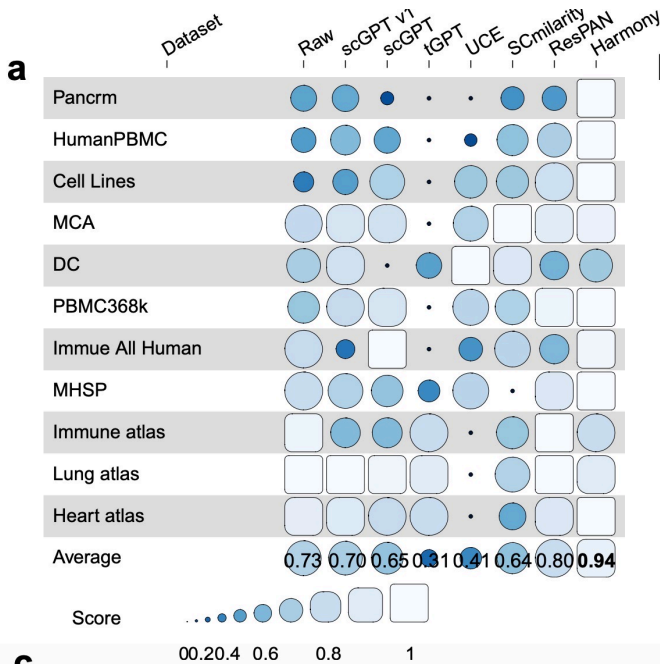
Bian, Xuegong Zhang. "scMulan: a multitask generative pre-trained language model for single-cell analysis." bioRxiv (2024): 2024-01.

Overall evaluations of scLLMs from scEval

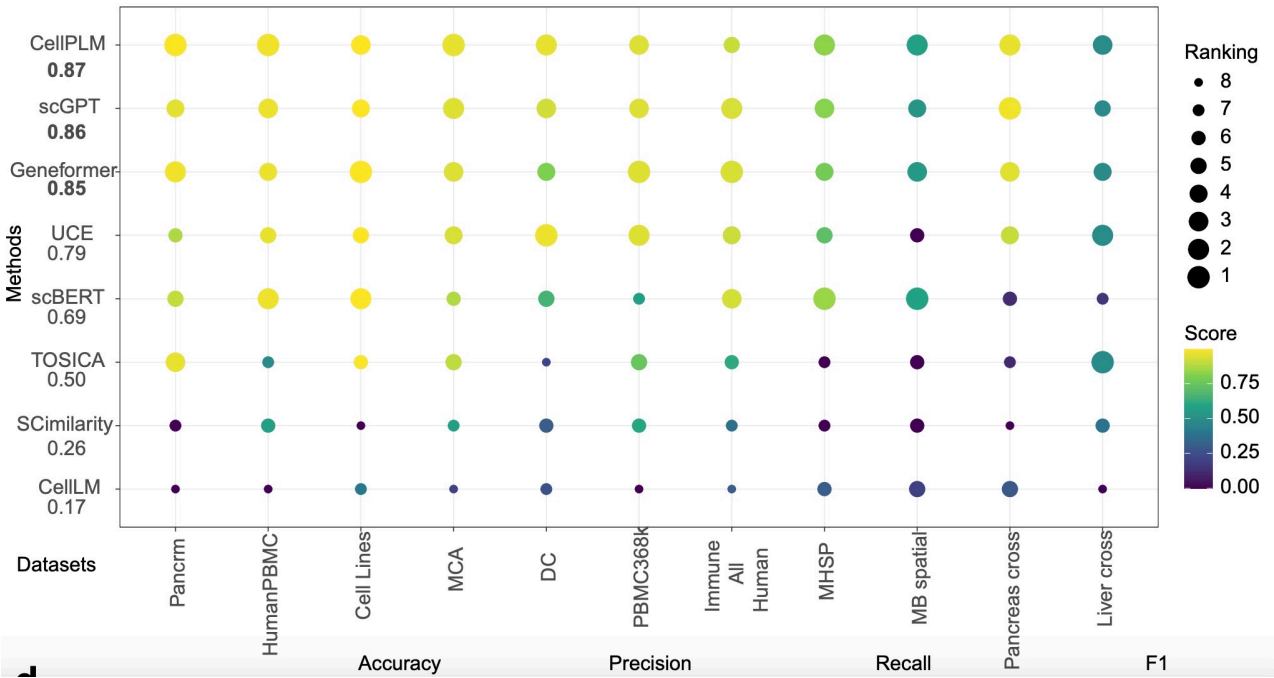


Tianyu Liu, Kexing Li, Yuge Wang, Hongyu Li, Hongyu Zhao. Evaluating the Utilities of Foundation Models in Single-cell Data Analysis. bioRxiv 2023.09.08.555192

Task evaluations of scLLM from scEval



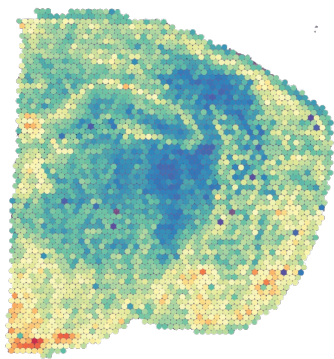
Batch correction



Cell type annotation

Spatial Foundation Models

Visium whole transcriptome

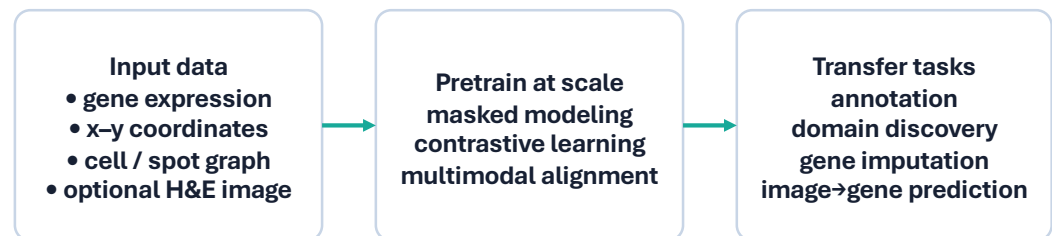
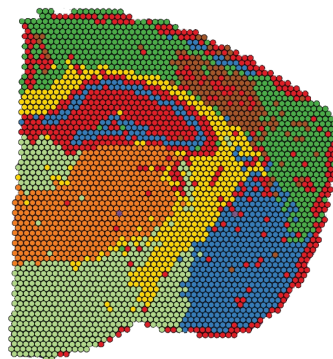


Total UMI



Example: spatial spots/clusters over a tissue section

Visium spot clusters



Core idea

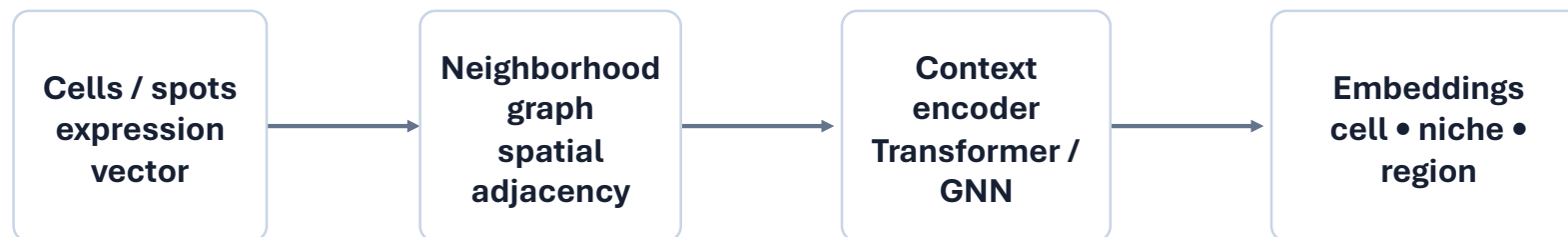
A spatial foundation model is pretrained on many spatial omics samples to encode cells/spots in context, then adapted or prompted for downstream analyses.

Why it matters: it can reduce per-study model training, improve cross-sample consistency, and expose tissue niches that are invisible in dissociated scRNA-seq.

Goal: learn reusable representations of cells, spots, niches and tissue regions so models transfer across organs, diseases and spatial platforms.

How models represent a tissue section?

Most spatial FMs convert tissue into tokens or graphs, then learn context-aware embeddings.



Token choices

gene-ranked “sentences” • binned genes
• spots • cells • image patches

Spatial context

kNN / radius graphs • tissue regions • niche-
level aggregation • SE(2)-aware geometry

Outputs

cell-type labels • domains • spatially
variable genes • predicted expression •
retrieval

Masked expression recovery
Predict hidden genes or
tokens

Contrastive spatial learning
Pull together nearby / matched
views

Image–gene alignment
Match H&E patches to
transcriptome

Hierarchical objectives
Cell $\rightleftharpoons$ niche $\rightleftharpoons$ tissue region

general embeddings

few-shot labels

domain maps

gene imputation

cross-modal query

Representative spatial transcriptomics foundation models

Model	Main modality	Core architecture	Typical outputs
Nicheformer	scRNA-seq + targeted spatial cells	Transformer trained on SpatialCorpus-110M	spatially aware cell embeddings; niche transfer
Novae	ST cell/spot graphs	Graph foundation model	zero-shot spatial domains; SVG/pathway analyses
OmiCLIP / Loki	H&E + Visium transcriptome	visual-omics contrastive model	alignment, annotation, deconvolution, image→gene
STPath	whole-slide images + ST	generative geometry-aware Transformer	zero-shot prediction of 38,984 genes
SToFM / HEIST	high-res ST graphs/multiscale data	SE(2) Transformer / hierarchical graph Transformer	region segmentation, annotation, imputation, clustering
scGPT-spatial / ST-Align	scGPT continual ST pretraining / image-gene pairs	Transformer alignment variants	cell typing, ST-specific embeddings, multimodal matching

Expression-and-neighborhood models: Nicheformer and Novae

Nicheformer

Transformer trained on both human/mouse single-cell and targeted spatial transcriptomics. SpatialCorpus-110M spans >110M cells and 73 organs/tissues.

Best fit: spatially aware cell embeddings, transfer of niche information, comparison of dissociated and spatial data.

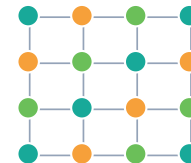


cell tokens with learned niche context

Novae

Graph-based foundation model for single-cell or spot-resolution ST. Designed for zero-shot spatial domain assignment across gene panels, tissues and technologies.

Best fit: spatial domain inference, batch correction, spatially variable genes/pathways, tissue architecture analysis.



spatial graph embeddings

Multimodal spatial FMs: connecting morphology to transcriptome

OmiCLIP / Loki

Uses paired H&E patches and Visium transcriptomics; OmiCLIP underlies Loki functions for tissue alignment, annotation, deconvolution, image-transcriptomics retrieval and ST gene-expression prediction.

STPath

Generative foundation model pretrained on WSIs paired with ST profiles. Reported to predict gene expression across 38,984 genes and 17 organs without downstream fine-tuning.



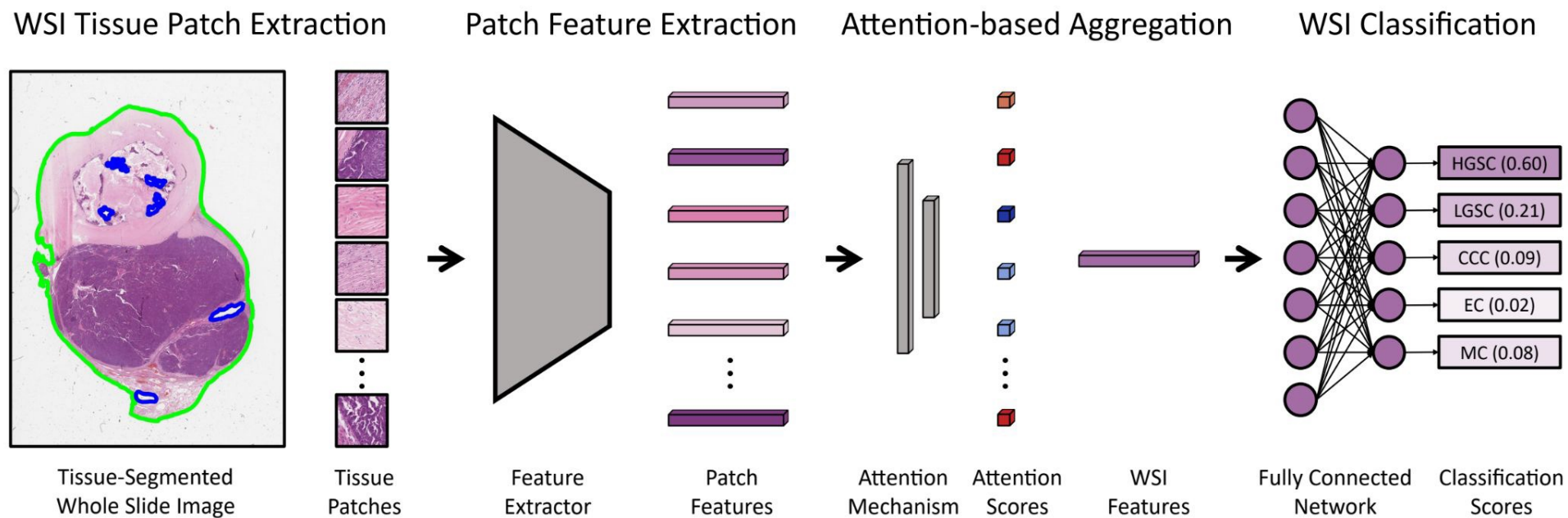
H&E Foundation Models



From morphology to embeddings

Large-scale pretraining turns routine H&E slides into general-purpose features for diagnosis, biomarker prediction, prognosis, retrieval, and multimodal reasoning.

H&E Foundation Models

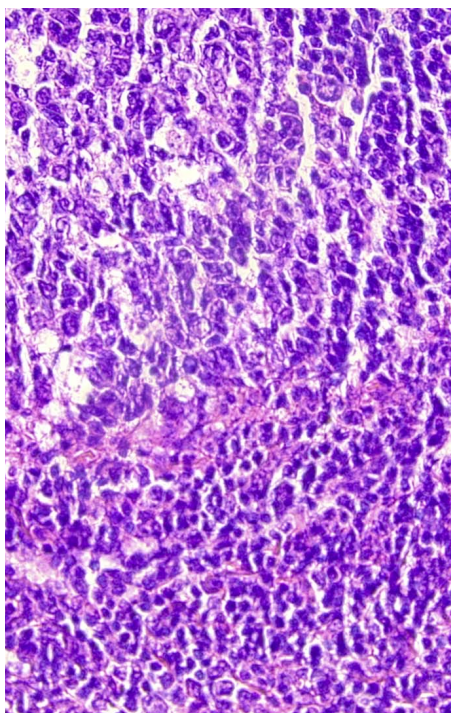


Jack Breen et al., npj Precision Oncology, 2025

Representative H&E / pathology foundation models

Model	Training signal	Scale / data	Best viewed as	Typical strengths
UNI / UNI2	Self-supervised vision	>100M H&E patches from >100k WSIs; UNI2 >200M H&E/IHC images	General-purpose patch encoder	Broad tissue generalization; strong feature extractor
CONCH	Vision-language contrastive	1.17M histopathology image-caption pairs	Patch-level V-L model	Retrieval, zero-shot, captioning, segmentation/classification
Virchow / Virchow2	DINOv2-style SSL	Virchow2: 3.1M WSIs; 632M / 1.85B params	Scaled patch encoder	High-performing morphology/biomarker features
Prov-GigaPath	Tile + whole-slide pretraining	1.3B tiles from 171,189 WSIs, 31 tissue types	Whole-slide encoder	Slide-level tasks; ultra-large-context modeling
H-Optimus-0	SSL vision	1.1B params; >500k H&E slides	Large open histology encoder	Strong open-source baseline at scale
TITAN / PRISM	Multimodal slide-report alignment	TITAN: 335,645 WSIs + reports + synthetic captions; PRISM uses Virchow embeddings + reports	Slide-level V-L models	Text prompting, report generation, zero/few-shot slide reasoning

UNI and CONCH



UNI

Self-supervised pathology model trained on >100M images from >100,000 diagnostic H&E WSIs across 20 major tissue types.

Designed as a general-purpose feature extractor for classification, retrieval, segmentation, and slide-level modeling.

Best fit: reusable visual embeddings

CONCH

Vision-language model trained with histopathology images, biomedical text, and >1.17M image-caption pairs.

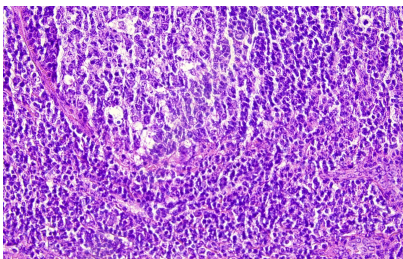
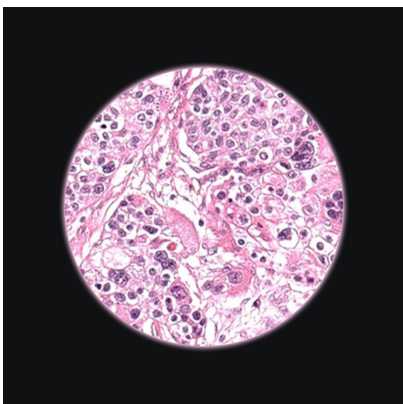
Adds text-to-image retrieval, image-to-text retrieval, captioning, zero-shot/few-shot classification, and interpretable prompting.

Best fit: image-text reasoning

Takeaway

UNI is useful when labels are available downstream; CONCH is useful when text, retrieval, or zero-shot behavior matters.

Scaling vision-only H&E encoders: Virchow2 and H-Optimus-0



Virchow2

Family of pathology ViTs trained with DINOv2-style, pathology-adapted self-supervision on 3.1M WSIs.

Published variants include Virchow2 (632M parameters) and Virchow2G (~1.85B parameters), with mixed-magnification training.

H-Optimus-0

Open-source histology ViT with ~1.1B parameters trained on >500,000 H&E whole-slide images.

Useful as a strong open baseline for H&E feature extraction, with active discussion around distillation and robustness.

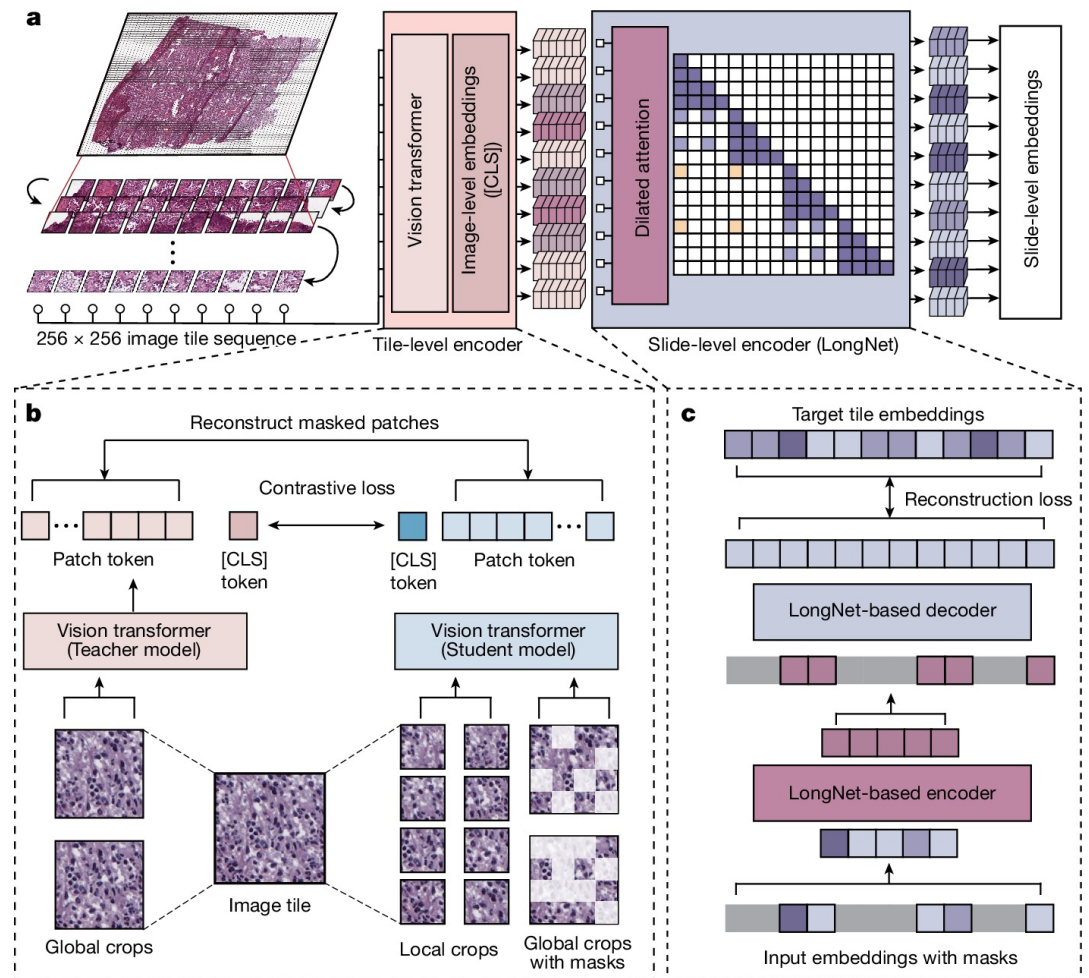
Practical lesson

Scale helps, but “best” depends on tissue type, task, label regime, WSI aggregation method, and domain shift. Recent benchmarks show top models are close, with CONCH/Virchow2 often leading across mixed tasks.

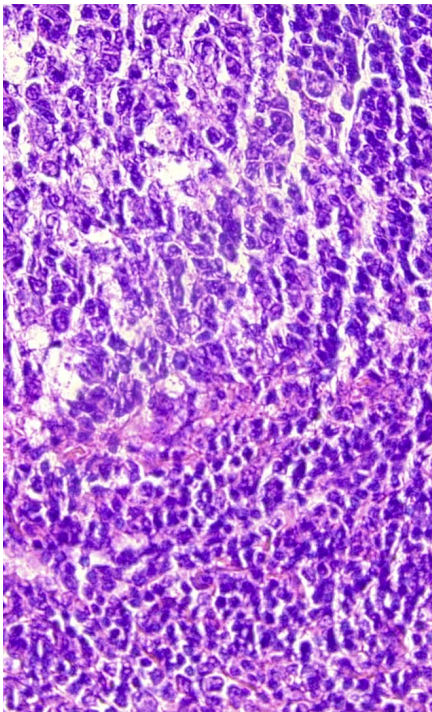
Prov-GigaPath

- Tile-encoder trained with DINOv2
- Slide-encoder trained with masked autoencoder
- Combines local tile representations with slide-level aggregation using a transformer architecture designed for gigapixel pathology.
- Trained on 171k WSIs from 30k patients

Source: Xu, A whole-slide foundation model for digital pathology from real-world data, 2024 (Microsoft)



Multimodal whole-slide models



TITAN

Transformer-based pathology Image and Text Alignment Network.

Pretrained on 335,645 WSIs using visual self-supervision plus report/caption alignment; uses 423,122 synthetic captions.

PRISM

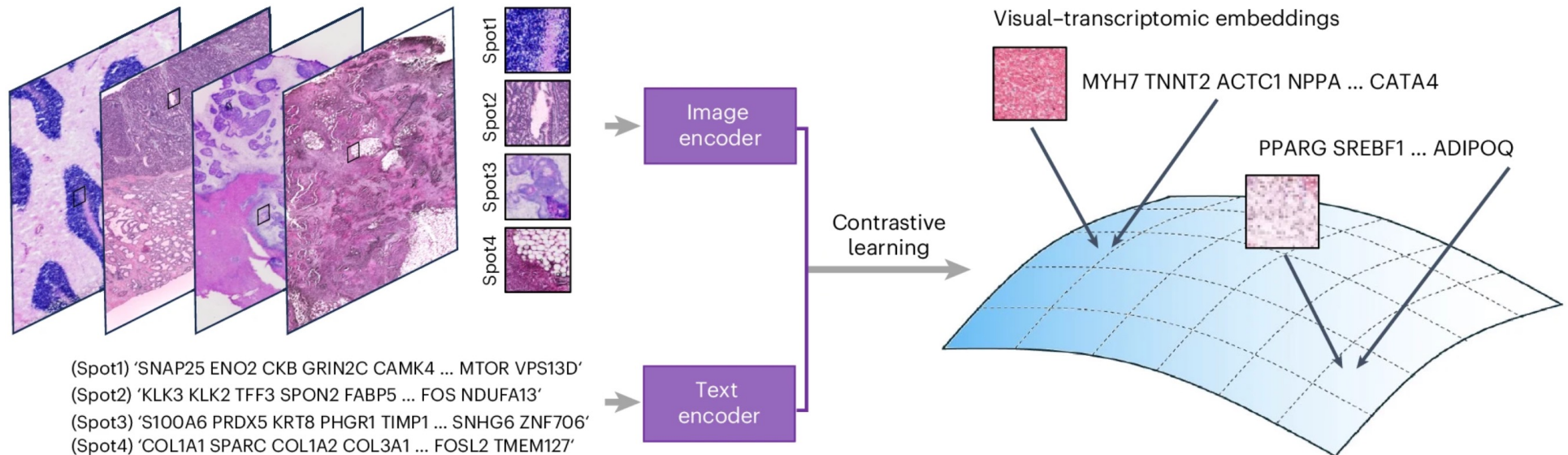
Slide-level multimodal generative model built on Virchow tile embeddings and clinical report text.

Supports slide embeddings, text prompting, zero-shot cancer detection/subtyping, and report-style generation.

Why it matters

Pathology is not only visual: diagnostic categories, uncertainty, rare entities, and biomarkers are often encoded in language. Multimodal FMs can connect morphology with clinical semantics.

OmiCLIP



- Visual-omics foundation model that integrates H&E with spatial transcriptomic profiles from Visium tissue patches.
- Transcriptomic profiles were converted into gene "sentences" by concatenating the top-expressed genes within each spatial patch.
- trained on a large-scale dataset of 2.2 million paired image-transcriptomics patches across 32 organs.
- Built on OmiCLIP, the Loki platform supports tissue alignment, tissue annotation, cell-type decomposition, image-transcriptomics retrieval, and gene expression prediction from H&E images.

Summary

- Foundation models for genes (transcriptome) and image (H&E)
- Spatial embedding
- May be better in signal-to-noise ratios, batch effects...
- Do not perform well on zero-shot
- Need adaptation and prompts to perform well
- Challenges in adaptation:
 - Compute intensive
 - Limited training data
 - Limited interpretability

Acknowledgments

This file is for the educational purpose only. Some materials (including pictures and text) were taken from the Internet at the public domain.